

Research on Anti-Windisturbance Technology for Quadrotor UAVs in Urban Rail Transit Tunnel

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Abstract—With the rapid development of urban rail transit, the demand for underground tunnel inspection is increasing day by day. Traditional manual inspection has low efficiency and high labor intensity, while drone inspection has the advantages of flexibility and efficiency. However, urban rail transit tunnel engineering is complex, and there are various strong wind disturbance scenarios in different tunnel environments, which seriously affect the stability and flight safety of drones. A double-layer gain adaptive sliding mode wind disturbance attitude control algorithm is proposed for the inner loop attitude angle control loop of unmanned aerial vehicles. By dynamically adjusting the inner loop control gain, the controller is always on the sliding mode control surface, which enhances the robustness of attitude control of unmanned aerial vehicles in turbulent random wind fields in tunnels with higher real-time performance and smaller error adjustment amplitude; By combining the outer loop PID control method, the accuracy of drone flight control is ensured. MATLAB simulation experiments show that compared with the traditional dual loop PID control strategy, this method can better suppress the influence of wind disturbance and improve the stability of drone flight; Through on-site testing in urban rail transit tunnel environments, the drone's flight status is stable, which can effectively ensure the reliability of drone flight in urban rail transit tunnel environments.

Keywords—urban rail transit; unmanned aerial vehicle inspection; wind disturbance control; sliding mode control

I. Introduction

As a core component of modern urban public transportation, the safety and stability of tunnel structures in urban rail transit systems are paramount. With extended operational years, tunnels become increasingly vulnerable to long-term impacts from train vibrations, groundwater seepage, and structural aging, potentially leading to cracks, leaks, and lining failures. Nighttime maintenance operations also risk equipment misplacement and cable trough cover misalignment, which could cause derailment hazards if not promptly addressed. Therefore, efficient tunnel inspection is essential

for operational safety. However, traditional inspection methods, such as primarily manual patrols or rail inspection vehicles, face three critical limitations [1].

- Inefficiency: Manual inspections require personnel to walk or operate low-speed rail inspection vehicles at speeds below 5 km/h, making comprehensive checks on multi-kilometer subway tunnels time-consuming.
- Physical strain: High-altitude inspections necessitate manual handling of ladders, posing significant operational challenges.
- Detection limitations: Human eyes struggle to detect subtle cracks, structural deformations, or loose components during aerial inspections, while reliance on subjective judgment often results in missed detections or misjudgments.

In recent years, drone technology has gained widespread application in power grid, bridge, and agricultural inspection fields due to its flexibility, high efficiency, rapid response, and capability to carry multiple sensors. Compared with traditional manual inspections, drone inspections offer advantages such as enhanced mobility, flexible deployment, and high-precision detection. By utilizing AI algorithms to automatically identify defects, they effectively reduce the workload of inspection personnel. Despite these significant advantages, urban rail transit tunnels present unique challenges. The tunnel's geographical location, structural characteristics, length, and equipment layout create strong wind disturbances as shown in Figure 1. Under normal conditions, wind speeds reach 5m/s ~ 10 m/s, with extreme sudden gusts exceeding this range, posing risks of flight attitude instability for drones [2]. Therefore, ensuring wind resistance capabilities in urban rail transit tunnels is a crucial prerequisite for implementing effective drone inspections.

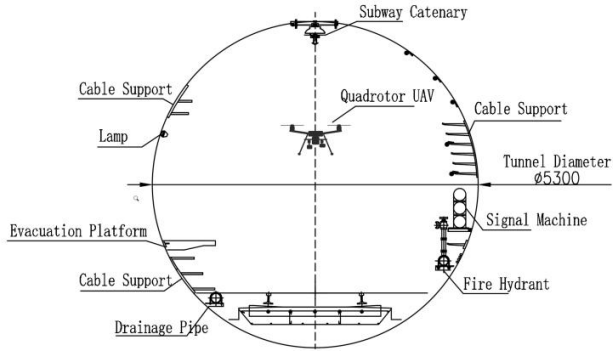


Figure 1 Schematic diagram of UAV inspection in typical urban rail tunnel

II. Dynamics modeling of quadrotor UAV

A. Basic dynamic model of quadrotor UAV

During tunnel inspection operations, quadcopters are typically employed due to spatial constraints in tunnel environments. While kinematic models focus on fundamental parameters like position, velocity, attitude, and angular velocity, dynamic models analyze motion dynamics and mechanical forces, incorporating the drone's mass and moment of inertia. This paper establishes the following modeling assumptions for quadcopters:

- The UAV is an axisymmetric rigid body with constant mass and moment of inertia, and the mass distribution is uniform;
- The center of gravity of the UAV is consistent with the geometric center;
- The thickness of the UAV is negligible, that is, the lift surface and the center of gravity generated by the rotation of the propeller are in the same plane;
- The curvature of the earth is negligible and the rotation speed is negligible.

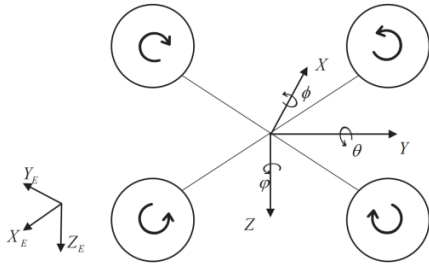


Figure 2 Quadrotor UAV model

According to [3], the dynamic equation of quadrotor UAV is as follows:

$$\begin{cases} \ddot{x} = -\frac{U_1}{m}(\sin \phi \sin \psi + \cos \phi \sin \theta \cos \psi) \\ \ddot{y} = -\frac{U_1}{m}(-\sin \phi \cos \psi + \cos \phi \sin \theta \sin \psi) \\ \ddot{z} = -\frac{U_1}{m} \cos \phi \cos \theta + g \end{cases} \quad (1)$$

$$\begin{cases} \ddot{\phi} = \frac{J_{yy} - J_{zz}}{J_{xx}} \dot{\theta} \dot{\psi} + \tau_{d\phi} + U_2 \\ \ddot{\theta} = \frac{J_{zz} - J_{xx}}{J_{yy}} \dot{\phi} \dot{\psi} + \tau_{d\theta} + U_3 \\ \ddot{\psi} = \frac{J_{xx} - J_{yy}}{J_{zz}} \dot{\phi} \dot{\theta} + \tau_{d\psi} + U_4 \end{cases} \quad (2)$$

$U_i (i = 1, 2, 3, 4)$ are the equivalent control variables for UAV position and attitude control, where U_1 is the equivalent synthesized lift, and $U_i (i = 2, 3, 4)$ are the equivalent control moment, corresponding to the roll, pitch, and yaw channels respectively. $X = (x, y, z)$ is the UAV's spatial position; m represents its mass; g indicates gravitational acceleration; and $\Theta = (\phi, \theta, \psi)$ stands for attitude angles corresponding to roll angle, pitch angle, and yaw angle. J_{xx}, J_{yy}, J_{zz} represent the inertia moments corresponding to the three axes, respectively. $\tau_{d\phi}, \tau_{d\theta}, \tau_{d\psi}$ respectively represent the external interference of the three channels. The four control inputs can be derived as:

$$\begin{cases} U_1 = k_T(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \\ U_2 = \frac{\sqrt{2}k_T l}{2J_{xx}}(-\omega_1^2 + \omega_2^2 + \omega_3^2 - \omega_4^2) \\ U_3 = \frac{\sqrt{2}k_T l}{2J_{yy}}(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \\ U_4 = \frac{K_M}{J_{zz}}(\omega_1^2 + \omega_2^2 - \omega_3^2 - \omega_4^2) \end{cases} \quad (3)$$

$\omega_i (i = 1, 2, 3, 4)$ represents the angular velocity of each motor; l represents the distance from the rotating center of each motor to the center of gravity of the UAV; k_T is the thrust coefficient; K_M is the torque coefficient.

B. Wind field model

During tunnel inspection flights, UAVs are inevitably disturbed by tunnel wind. Railway tunnels typically contain three types of wind fields: natural wind, piston wind, and fan-limited jet wind. According to wind field definitions, tunnel environments consist of mean wind fields, turbulent wind fields, and shear wind fields. During routine UAV inspections, which occur during the ventilation period when train operations are absent, the probability of shear wind fields is relatively low. The wind field environment exhibits superposition characteristics, allowing simulation of actual conditions by combining multiple wind fields. During flight, the high-speed rotation of quadcopter propellers generates wind fluctuations in tunnel environments, which are essentially the synthesis of turbulent and mean wind fields. As documented in [4], the resistance and moments experienced by UAVs during tunnel operations can be defined as:

$$\begin{cases} F_{\omega x} = C\rho S_x W_x^2 / 2 \\ F_{\omega y} = C\rho S_y W_y^2 / 2 \\ F_{\omega z} = C\rho S_z W_z^2 / 2 \end{cases} \quad (4)$$

$$\begin{cases} M_{\omega x} = C\rho S_x l W_x^2 / 2 \\ M_{\omega y} = C\rho S_y l W_y^2 / 2 \\ M_{\omega z} = C\rho S_z l W_z^2 / 2 \end{cases} \quad (5)$$

Where: C is the drag coefficient; ρ is the air density; and S_x, S_y, S_z are the projected area of the UAV on three planes.

III. Double gain adaptive sliding mode control anti-disturbance strategy

Based on the four-rotor UAV dynamics model, Equation (1) represents the control loop for position information, while Equation (2) governs the attitude control loop. This allows the UAV controller to be designed with two integrated loops, as illustrated in Figure 2. In tunnel environments with random wind fields, the outer loop employs a PID controller to suppress persistent disturbances and gradual random disturbances, ensuring precise position control. Meanwhile, turbulent wind fields primarily introduce sudden strong disturbances, requiring the inner-loop attitude controller to respond rapidly. For UAVs that demand rapid real-time response and high robustness, sliding mode control demonstrates superior performance [5]. This paper proposes a dual-gain adaptive sliding mode controller for the UAV's inner-loop control system, and the system structure is shown in Figure 3.

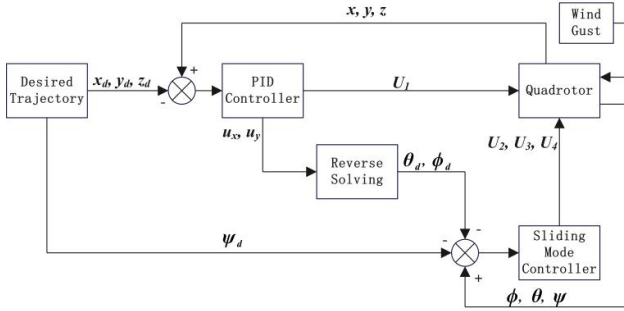


Figure 3 Schematic diagram of four-rotor UAV control system

In order to simplify the calculation model, a set of virtual control quantities are defined according to Equation (1) as shown in Equation (6), where: U_x, U_y, U_z are three virtual input control quantities of the position control loop, and can be solved for thrust U_1 and attitude Angle ϕ_d and θ_d , as shown in Equation (7).

$$\begin{cases} U_x = -U_1(\sin \phi \sin \psi + \cos \phi \sin \theta \cos \psi)/m \\ U_y = -U_1(-\sin \phi \cos \psi + \cos \phi \sin \theta \sin \psi)/m \\ U_z = -U_1 \cos \phi \cos \theta /m + g \end{cases} \quad (6)$$

$$\begin{cases} U_1 = m \sqrt{U_x^2 + U_y^2 + (g - U_z)^2} \\ \phi_d = \sin^{-1} \frac{-m(U_x \sin \psi_d - U_y \cos \psi_d)}{U_1} \\ \theta_d = \tan^{-1} \frac{U_x \cos \psi_d + U_y \sin \psi_d}{U_z - g} \end{cases} \quad (7)$$

Based on the system architecture shown in Figure 2, an adaptive sliding mode controller is designed in the UAV's inner-loop control loop to achieve precise flight attitude tracking under wind disturbances. Taking roll angle as an example, the tracking error is defined as $e_\phi = \phi - \phi_d$. According to the dynamics equation (1), the design sliding

mode surface can be expressed as:

$$s_\phi = \dot{e}_\phi + c e_\phi \quad (8)$$

Here, c is a constant, and the derivative of the sliding surface is obtained as:

$$\begin{aligned} \dot{s}_\phi &= \ddot{e}_\phi + c \dot{e}_\phi \\ &= U_2 + \frac{J_{yy} - J_{zz}}{J_{xx}} \dot{\phi} \dot{\psi} + \tau_{d\phi} - \ddot{\phi}_d + c \dot{e}_\phi \end{aligned} \quad (9)$$

For this system, the control input is designed as follow:

$$U_2 = -c \dot{e}_\phi - \frac{J_{yy} - J_{zz}}{J_{xx}} \dot{\phi} \dot{\psi} + \ddot{\phi}_d - k(t) \text{sgn}(s_\phi) \quad (10)$$

During the flight of the UAV in the tunnel, the wind disturbance is generally continuous. In this case, it is assumed that the wind disturbance $\tau_{d\phi}$ is bounded and continuously differentiable, and there exists an unknown bounded constant a_0 and a_1 to satisfy:

$$\begin{cases} \tau_{d\phi} < a_0 \\ \dot{\tau}_{d\phi} < a_1 \end{cases} \quad (11)$$

According to the slip surface formulas (8) and (9) and the controller input (10), the derivative of the Lyapunov function of the controller $V = s_\phi^2/2$ can be obtained as follows:

$$\begin{aligned} \dot{V} &= s_\phi \cdot \dot{s}_\phi \\ &= s_\phi \cdot \left(U_2 + \frac{J_{yy} - J_{zz}}{J_{xx}} \dot{\phi} \dot{\psi} + \tau_{d\phi} - \ddot{\phi}_d + c \dot{e}_\phi \right) \\ &= s_\phi \cdot [\tau_{d\phi} - k(t) \text{sgn}(s_\phi)] \\ &\leq -|s_\phi| \cdot [k(t) - |\tau_{d\phi}|] \end{aligned} \quad (12)$$

When the switching control $k(t) > |\tau_{d\phi}|$ gain is bounded, the sliding mode surface of the controlled system can converge to zero within a finite time, and the system has asymptotic stability. Combined with Equation (10), the switching function is defined $v(t) = -[k(t) + \eta] \text{sgn}(s_\phi)$, where η is a normal number "sufficiently small" and $k(t)$ is a variable satisfying adaptive control.

Based on the equivalent control principle, when $s_\phi = \dot{s}_\phi = 0$, we can obtain the $v(t) = -\tau_{d\phi}$ at this stage. To ensure the system reaches the sliding mode surface and initiates sliding mode motion, the equivalent control quantity $v_{eq}(t) = -\tau_{d\phi}$ is maintained to sustain this motion. Once the system achieves successful sliding mode movement, the absolute value of the equivalent control quantity becomes equal to that of the disturbance quantity. Subsequently, through estimation $v_{eq}(t)$ using a first-order low-pass filter, the approximate estimated quantity $\bar{v}_{eq}(t)$ is derived as:

$$\dot{\bar{v}}_{eq}(t) = [(-k(t) + \eta) \text{sgn}(s_\phi) - \bar{v}_{eq}(t)]/b \quad (13)$$

Here, b is the low-pass filter time constant. To solve $k(t)$, we redefine an error variable $\delta(t)$ as:

$$\delta(t) = k(t) - \frac{1}{\alpha} |\bar{v}_{eq}(t)| - \varepsilon \quad (14)$$

Here, $0 < \alpha < 1, \varepsilon > 0$. When $\delta(t) = k(t) =$

$\frac{1}{\alpha}|\bar{v}_{eq}(t)| - \varepsilon > |\bar{v}_{eq}(t)| = |\tau_{d\phi}|$, the sliding motion can be maintained. At this time, the control gain $k(t)$ is designed to satisfy the condition $\delta \rightarrow 0$, which can be defined as:

$$\dot{k}(t) = -\rho(t)sgn(\delta(t)) \quad (15)$$

$\rho(t)$ is a varying scalar that represents the upper limit of the rate of disturbance change. It is assumed to have the following form:

$$\rho(t) = r(t) + r_0 \quad (16)$$

Where, r_0 is a normal number. According to Equation (7), in order to maintain the sliding mold movement, $r(t)$ is necessary to adaptively adjust itself to ensure $r(t) + r_0 > |\tau_{d\phi}|$. The designed adaptive control rate $r(t)$ is:

$$\dot{r}(t) = \begin{cases} \gamma|\delta(t)|, & \text{when } |\delta(t)| > \delta_0 \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

Here, $\gamma > 0, \delta_0 > 0$. Meanwhile, under the condition $\dot{\tau}_{d\phi} < a_1$, if for any given a_1 and δ_0 , it can be solved according to formula (9) as follows:

$$\frac{1}{4}\varepsilon^2 > \delta_0^2 + \frac{1}{\gamma}\left(\frac{qa_1}{\alpha}\right)^2 \quad (18)$$

According to the designed control input formula (10) and equations (15) and (16), the $|\delta(t)|$ holding can be made smaller than $\varepsilon/2$ in a finite time, so as to ensure that the sliding motion can be sustained. In addition, $\rho(t)$ and $k(t)$ can always remain within a limited range.

If $|\delta(t)| < \varepsilon/2$, it can be obtained according to Equation (14)

$$|\delta(t)| = \left| k(t) - \frac{1}{\alpha}|\bar{v}_{eq}(t)| - \varepsilon \right| < \varepsilon/2 \quad (19)$$

When $k(t) > \frac{1}{\alpha}|\bar{v}_{eq}(t)| + \frac{\varepsilon}{2} > |\bar{v}_{eq}(t)| = |\tau_{d\phi}|$, the system satisfies the sliding mode motion condition.

IV. Simulation

To evaluate the performance of the sliding-mode controller in wind field environments, we incorporated external wind disturbances simulated by the Dryden Wind Farm Model into the dynamic model input of the UAV system. The parameters of the Dryden Wind Farm Model are specified $W = 10m/s$. By setting the simulation duration $T = 60s$, the external wind field was introduced at 5s, and the wind field acting on the UAV is illustrated in Figure 4.

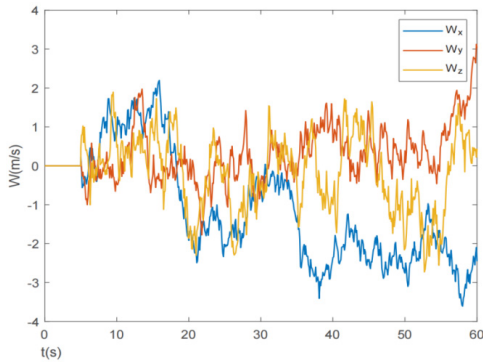


Figure 4 Stream wind field model

Under the wind field conditions, MATLAB simulation is used to verify the robustness and anti-interference performance of the proposed sliding mode controller. Table 1 shows the dynamic parameters of the quad-rotor UAV system.

Table 1 Four-rotor UAV parameter table

Parameter	Description	value	Unit
m	Mass of quadrotor	0.25	kg
l	The distance between the center of gravity of the UAV and the motor shaft	0.125	m
J_{xx}	Rotational inertia of the UAV around the x-axis	0.003	kg·m ²
J_{yy}	Rotational inertia of UAV around y-axis	0.003	kg·m ²
J_{zz}	Rotational inertia of UAV around z axis	0.006	kg·m ²
S_x	The projection area of the UAV on the y-z plane	0.1	m ²
S_y	The projection area of the UAV on the x-z plane	0.1	m ²
S_z	The projection area of the UAV on the x-y plane	0.2	m ²

Meanwhile, the outer-loop PID parameters are set as: $k_{xp} = k_{yp} = 5$, $k_{zp} = 2$, $k_{xi} = k_{yi} = 0.02$, $k_{zi} = 0$, $k_{xd} = k_{yd} = 3.8$, $k_{zd} = 2.5$. The attitude sliding mode control parameters for the roll angle are $r_0 = 0.5$, $\alpha = 0.49$, $\delta_0 = 0.01$, $\varepsilon = 0.15$, $\gamma = 200$, $b = 0.01$, $c = 0.5$. The control parameters for pitch and yaw angles are identical to those of the roll angle. The drone's initial position is set at $X_0 = (1, 1, 2.5)$, with an initial attitude angle of $\Theta = (0, 0, 0)$. For comparative analysis, two control methods are simulated:

- Method 1: Dual loop PID control;
- Method 2: Outer loop PID control, inner loop adopts sliding mode control.

Figure 5 presents tracking control results for three inner-loop attitude angles under two control methods. While Method 1 achieves tracking capability, it exhibits significant control quantity deviations during closed-loop adjustment with prolonged convergence times. The maximum deviations for roll angle, pitch angle and yaw angle were approximately 0.11rad, 0.05rad and 0.03rad respectively. In contrast, Method 2 employs sliding mode control, achieving maximum deviations of approximately 0.02rad, 0.01rad and 0.02rad for the same parameters, while demonstrating significantly shorter control quantity convergence time.

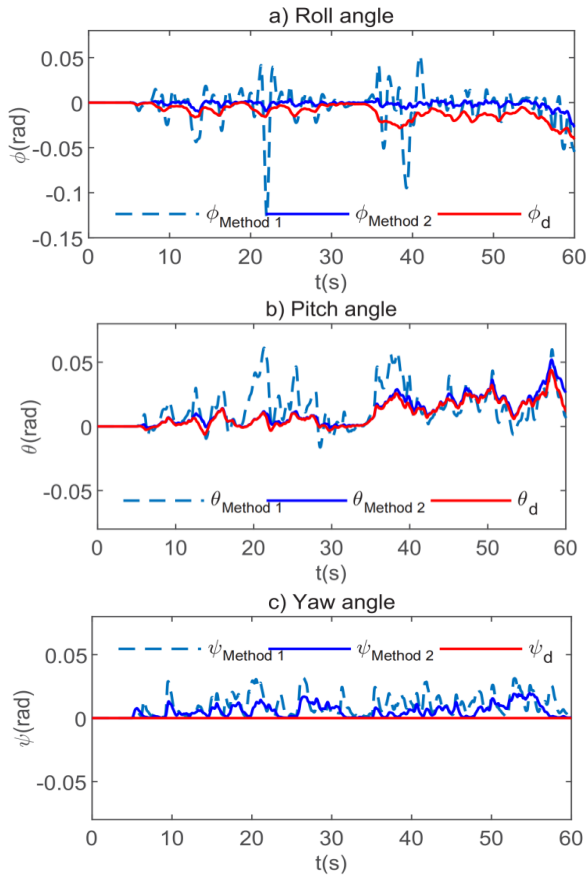


Figure 5. UAV attitude Angle control information

Figure 6 shows the position tracking control results of the outer loop under two control methods. It can be seen that under the inner loop sliding control with shorter response time and better control effect, the position control accuracy and tracking effect of the outer loop in method 2 are significantly better than that of method 1.

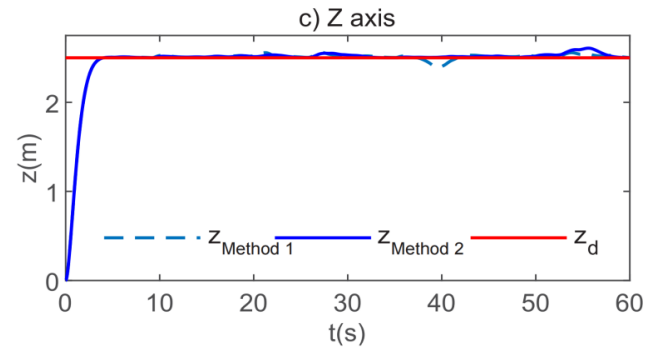
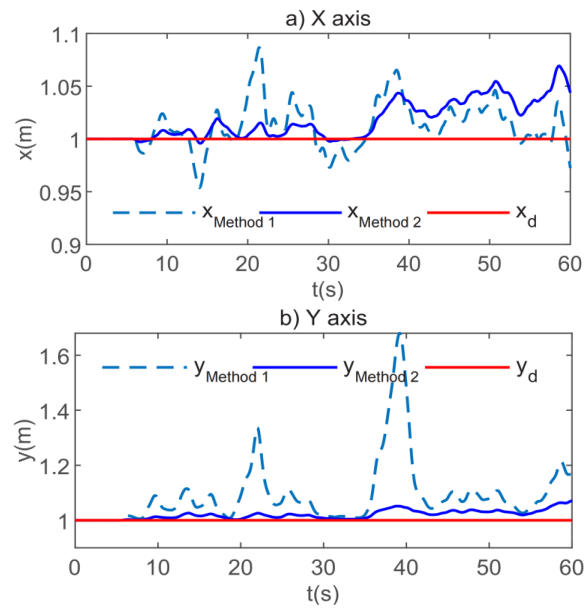


Figure 6. Position control information of UAV

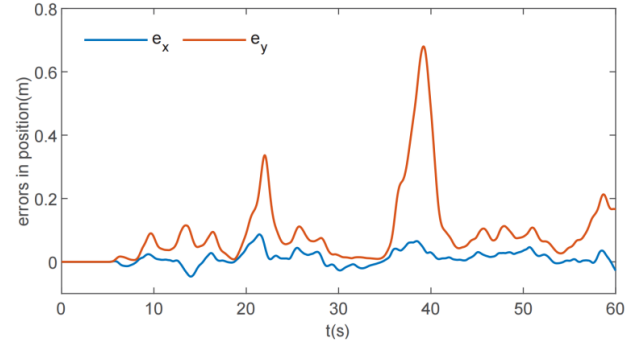


Figure 7. X-Y axis position error under method 1 control

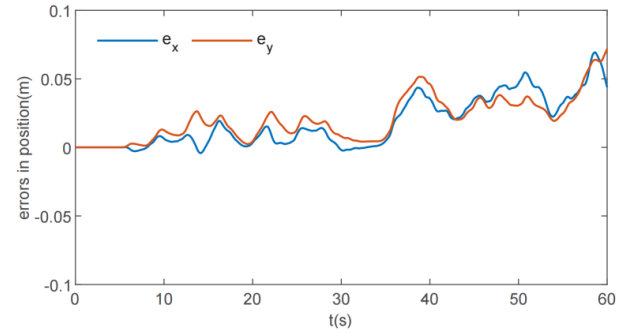


Figure 8. X-Y axis position error under method 2 control

Figure 7 and Figure 8 show the displacement error of the UAV in the x-y plane during the wind disturbance control process. The maximum displacement error of method 1 is about $0.7m$; the maximum displacement error of method 2 is only $0.06m$, and the control effect is better.

Path	Total time (s)
> FTES0	0.012
BLFSMC	0.005
BLFSMC1	0.004
PSBF_phitest	0.002
PSBF_thetatest	0.001
PSBF_psitest	0.001

Figure 9. Computational costs of the inner-loop sliding mode control

Additionally, we test the computational costs of the inner-loop sliding mode control method. The time consumption of the SMC is about $5ms$, as in Figure 9, which can meet the real-time requirements of the real-time tunnel applications.

Through comparative analysis, it is evident that the performance of the inner-loop attitude control system plays a crucial role in the overall flight control performance of the UAV. By employing inner-loop attitude sliding mode control technology, the adverse effects of random wind disturbances on the UAV can be more effectively mitigated. The system achieves smaller disturbances in the three attitude angles of the UAV, not only reducing tracking errors but also enabling rapid and precise stabilization of flight attitude, thereby better ensuring spatial positioning accuracy. Compared with traditional inner/outer-loop PID controllers, the inner-loop attitude sliding mode controller demonstrates superior performance and stability in UAV. Simulation results confirm that this method can effectively stabilize the flight state of UAV in external turbulent wind fields.

V. Real Environment Verification

Figure 10 shows the stability test of unmanned aerial vehicle hovering in a real urban rail tunnel environment. Due to the nighttime operation and maintenance window period of the experiment, there is no piston wind caused by train operation in the tunnel; The on-site wind environment is a combination of natural mean wind field and random turbulent wind field caused by intermittent tunnel fans, and the actual value of wind disturbance cannot be accurately measured. The UAV drone took off in the wind field of the tunnel and slowly rose to a height of 2.5 meters while hovering. The entire testing process lasted for one minute. From Figures 10 and 11, it can be seen that wind disturbance has a significant impact on the stability of the aircraft.



Figure 10 Field test in urban rail transit tunnel environment

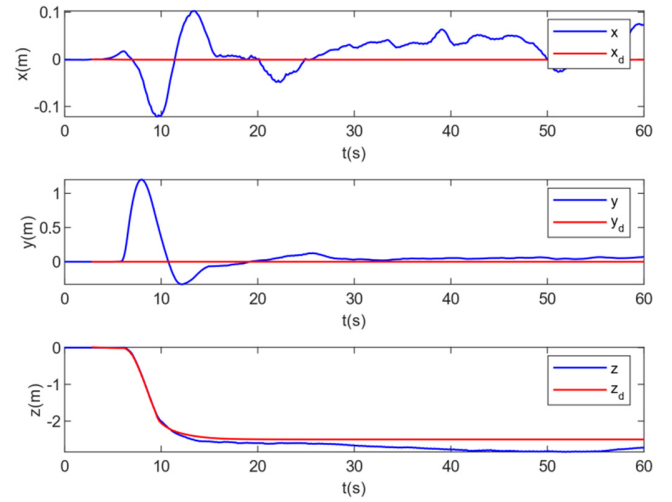


Figure 11 Position tracking curve of fixed-point hovering

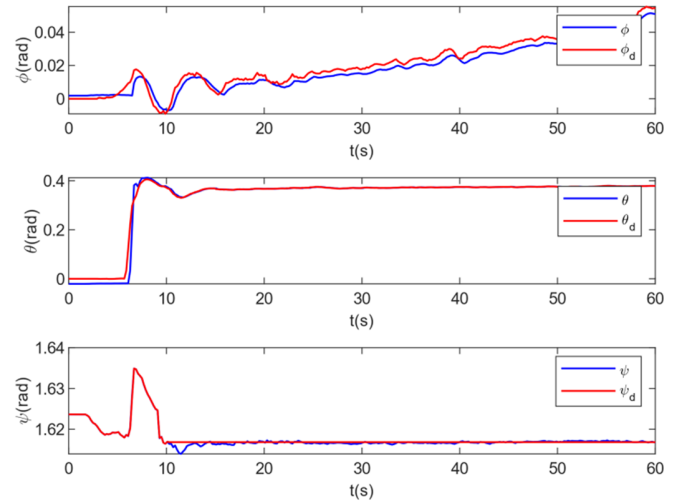


Figure 12 Attitude angle tracking curve of fixed-point hovering

From Figure 11, it can be seen that due to the limitations of the tunnel structure conditions, the turbulent airflow field inside the tunnel mainly runs along the tunnel direction. Therefore, the hovering position of the drone fluctuates within a certain range under the action of the wind field. The fluctuation of the x-axis in the same direction as the wind direction is particularly evident. After the expected attitude calculated by the position loop is transmitted to the attitude loop, the attitude controller adjusts the drone attitude in real time. Due to the forward thrust generated by the wind field on the drone, the calculated expected pitch angle is a positive value. Due to the constant need for unmanned aerial vehicles to adjust their attitude in the wind field, pitch angle adjustment is particularly noticeable, as shown in Figure 12. The information source of drone flight altitude comes from the laser rangefinder at the bottom of the aircraft. When the drone attitude changes, it will cause the actual altitude to deviate slightly from the predetermined altitude. Observing the expected curve and current position curve, it can be seen that the position and attitude errors of the drone are relatively

small, and the control system has good robustness to the random wind field in the testing environment.

VI. Conclusion

This article proposes an adaptive sliding mode attitude control method based on traditional sliding mode controllers to address the problem of turbulent wind disturbance commonly encountered during unmanned aerial vehicle tunnel inspections. The method ensures that the sliding mode control gain tracks the disturbance term in the absence of unknown disturbance information, while avoiding overestimation of switching gain, ultimately ensuring the continuous and stable control of the system. Through MATLAB simulation, compared with the internal and external dual loop PID control method, this scheme has better anti-interference ability and response performance. Although the on-site testing scenario in this article has certain limitations and cannot cover all wind disturbance conditions in tunnel environments, the testing in real tunnel environments proves the effectiveness of the proposed dual gain adaptive sliding mode control anti-interference strategy, providing a good foundation, theoretical verification, and technical support for further research and development of autonomous positioning, obstacle avoidance flight, and other inspection functions of tunnel inspection drones.

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